

Object Oriented Programming

Java



Swing in Java

- Swing is a **GUI widget toolkit** for Java.
- It is part of **Oracle's Java Foundation Classes (JFC)** – an API for providing a graphical user interface (GUI) for Java programs.
- Swing was developed to provide a more sophisticated set of GUI components than the earlier Abstract Window Toolkit (AWT).
- Unlike AWT components, Swing components are **platform-independent**.



AWT vs Swing

Java AWT	Java Swing
AWT components are platform-dependent .	Java swing components are platform-independent .
AWT components are heavyweight .	Swing components are lightweight .
AWT doesn't support pluggable look and feel .	Swing supports pluggable look and feel .
AWT provides less components than Swing.	Swing provides more powerful components such as tables, lists, scrollpanes, colorchooser, tabbedpane etc.
AWT doesn't follows MVC (Model View Controller)	Swing follows MVC .



Pluggable Look & Feel



MVC in Swing

The MVC pattern is a model of how a user interface can be structured. Therefore it defines the following 3 elements:

1. **Model** that represents the data for the application. **TableModel** is used for collecting the data.
2. **View** that is the visual representation of that data. **JTable** is a good example for the view.
3. **Controller** that takes user input on the view and translates that to changes in the model. A controller is normally implemented by an **ActionListener** or **Action**.

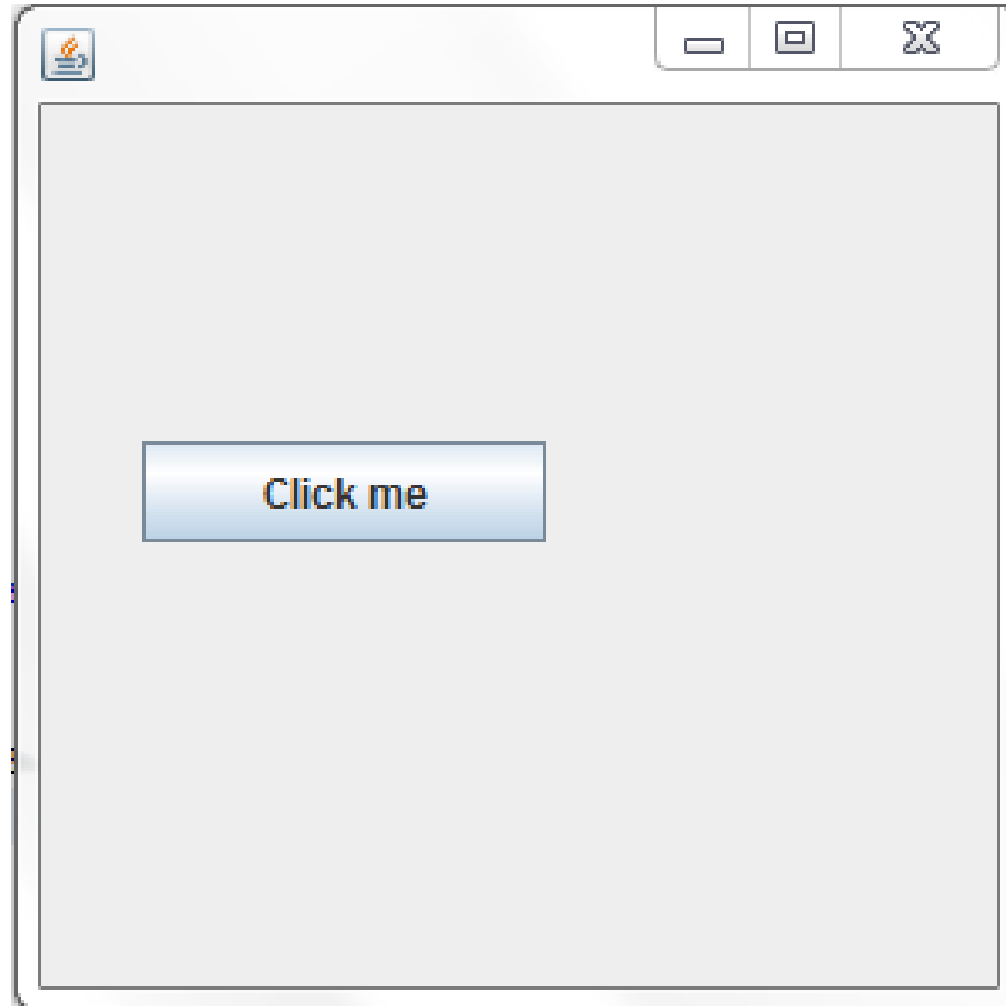


Creating Frame

```
import javax.swing.*;
public class First extends JFrame{
    public First(){
        JButton b=new JButton("Click me");
        b.setSize(80, 30);
        b.setLocation(30, 100); // setting button position
        add(b);           //adding button into frame
        setSize(300,300); //frame size 300 width and 300 height
        setLayout(null);   //no layout manager
        setVisible(true); //now frame will be visible, by default not visible
    }
    public static void main(String args[]){
        First f=new First();
    }
}
```



Creating Frame



JFileChooser With ActionListener

```
import javax.swing.*;  
import java.awt.event.*;  
public class First extends JFrame implements ActionListener{  
    JMenuBar mb;  
    JMenu file;  
    JMenuItem open;  
    JTextArea ta;
```

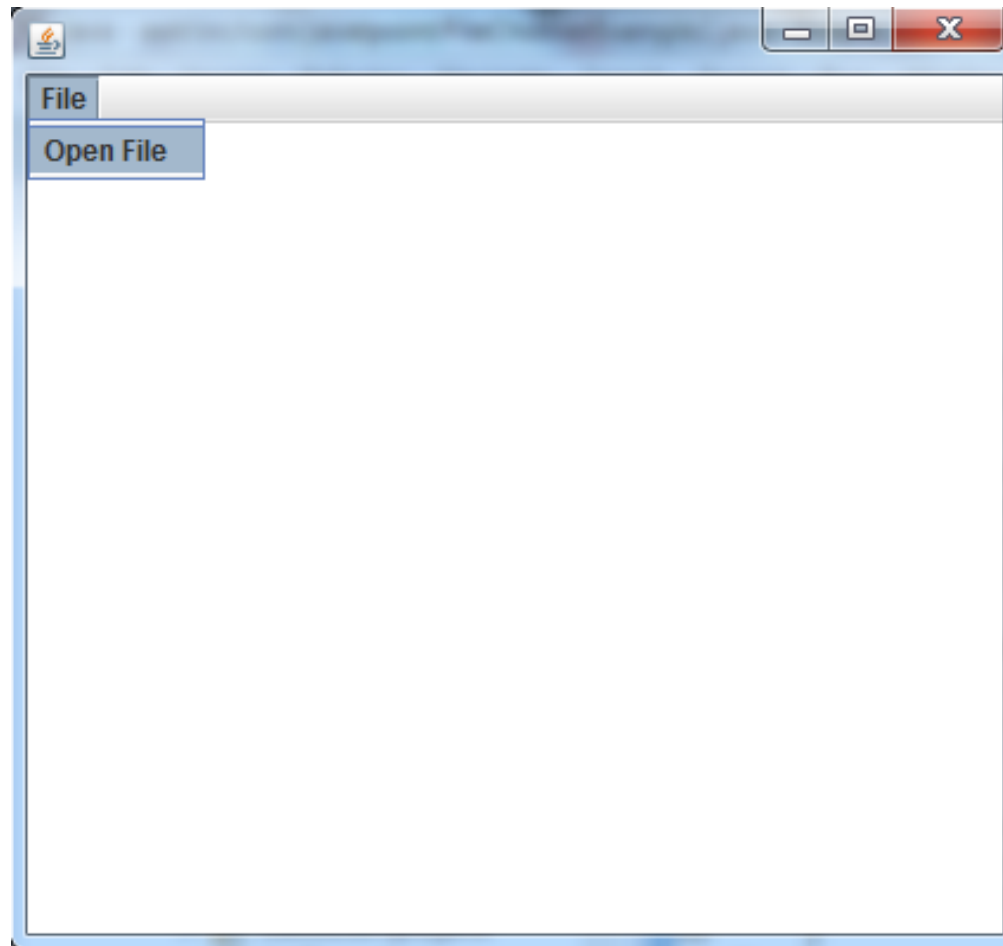


JFileChooser With ActionListener

```
public First(){  
    open = new JMenuItem("Open File");  
    open.addActionListener(this);  
    file = new JMenu("File");  
    file.add(open);  
    mb = new JMenuBar();  
    mb.setBounds(0,0,800,20);  
    mb.add(file);  
    ta = new JTextArea(800,800);  
    ta.setBounds(0,20,800,800);  
    add(mb);  
    add(ta);  
}
```



JFileChooser With ActionListener

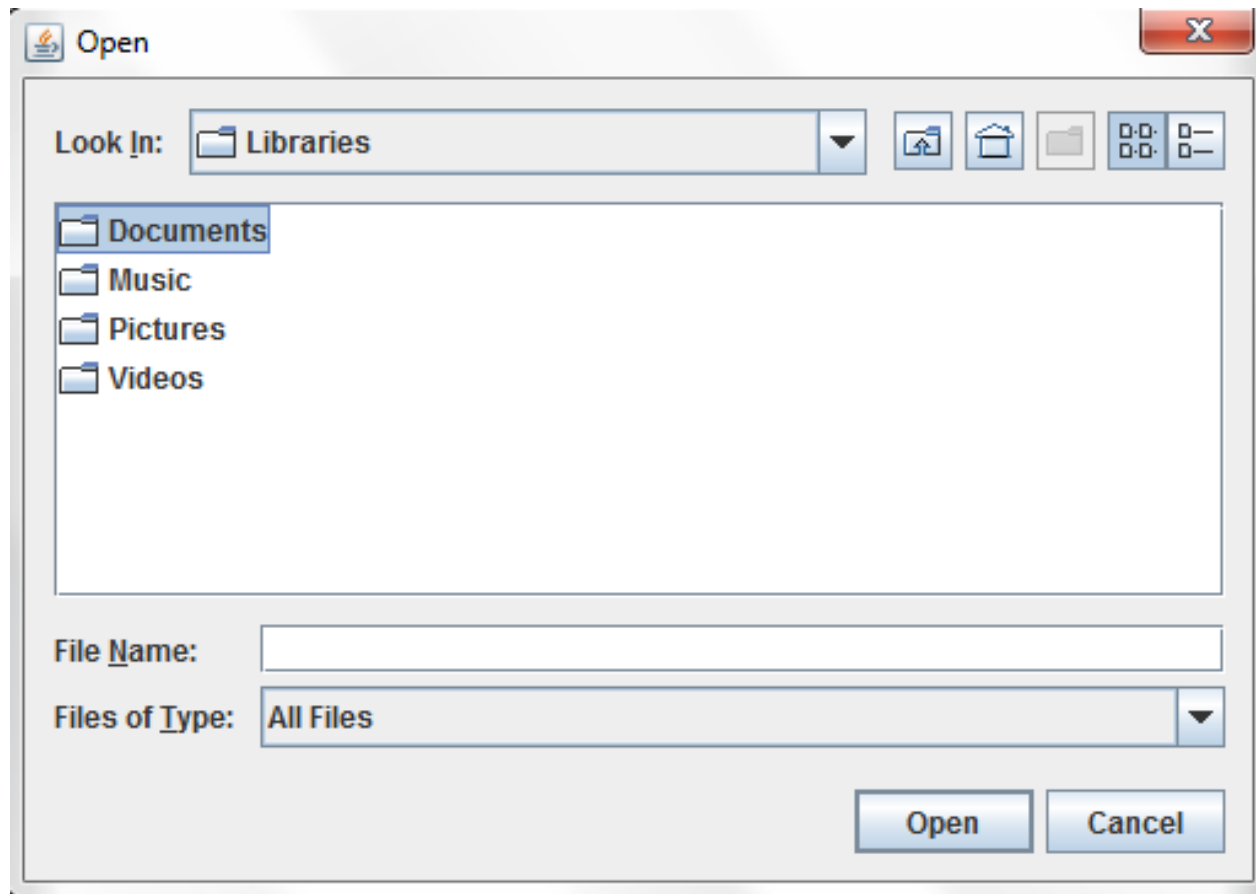


JFileChooser With ActionListener

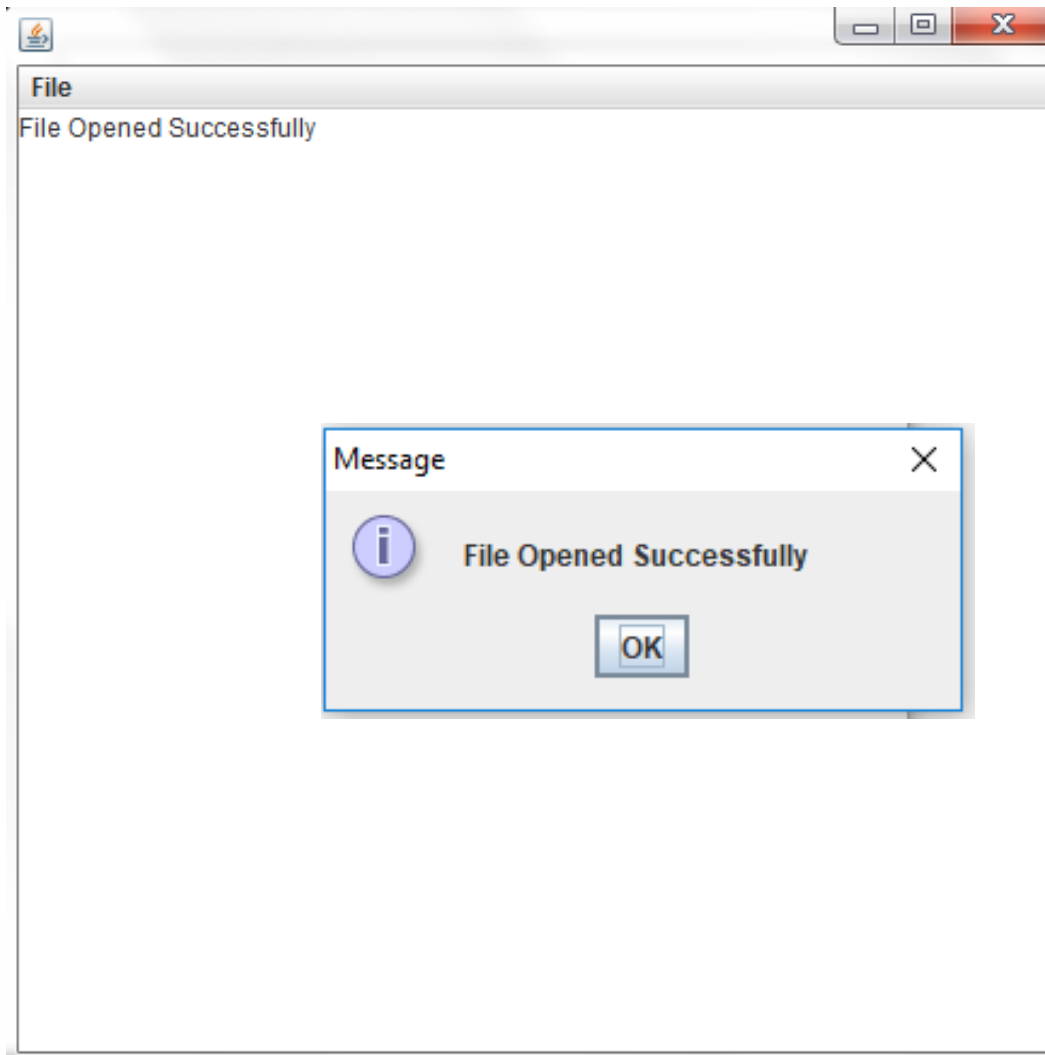
```
public void actionPerformed(ActionEvent e) {  
    String s2="";  
    if(e.getSource() == open){  
        JFileChooser fc = new JFileChooser();  
        int i = fc.showOpenDialog(this);  
        if(i == JFileChooser.APPROVE_OPTION){  
            s2 ="File Opened Successfully\n";  
            JOptionPane.showMessageDialog(null, s2);  
            ta.setText(s2);  
        }  
    }  
}
```



JFileChooser With ActionListener



JFileChooser With ActionListener



JFileChooser With ActionListener

```
public static void main(String[] args) {  
    First om = new First();  
    om.setSize(500,500);  
    om.setLocation(300, 100);  
    om.setLayout(null);  
    om.setVisible(true);  
    om.setDefaultCloseOperation(JFrame.EXIT_ON_C  
LOSE);  
}  
}
```

